

Effects of IMD on *A. caliginosa*

Protocol

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Objective : Construction of dose-response curves for growth and reproduction.

1 Generalities

1.1 Earthworms

Earthworms for these preliminary tests are *A. caliginosa* individuals from (or bred from individuals) fetch in a meadow in Versailles (Les Closeaux).

1.2 Design

Cosms are made with 200 g of Closeaux soil and earthworms are fed with 3 g/ind/14d.

Each modality is made with 5 replicates for growth and 5 replicates for reproduction.

The growth test lasts 28 days and the reproduction test lasts 28 days with a monitoring of cocoons of 28 days after the test.

For the growth test, one replicate correspond to 1 individuals per cosm. For the reproduction test, one replicate correspond to 2 individuals per cosm.

Table 1.1: Tested concentrations (mg/kg)

Tested concentrations (mg/kg)
0.0000
0.0016
0.0050
0.0158
0.0500
0.1581
0.5000
1.5811
5.0000

We need :

- 45 juveniles for the growth experiment,
- 90 adultes for the growth experiment.

1.3 Soil and horse dung needs

We need :

- 18 kg of Closeaux soil,
- 810 g of horse dung.

2 Protocol

2.1 Calculations

Table 2.1: Soil preparation for each IMD concentration

Concentration (mg/kg)	Q_{IMD} (mg)	Q_{tot} soil (g)	V_{eau} (CRE 70%)	V_{prim} (mL)
0.000000	0.000000	2200	602.756	0.0000
0.001581	0.003479	2200	602.756	0.0174
0.005000	0.011000	2200	602.756	0.0550
0.015811	0.034785	2200	602.756	0.1739
0.050000	0.110000	2200	602.756	0.5500
0.158114	0.347851	2200	602.756	1.7393
0.500000	1.100000	2200	602.756	5.5000
1.581139	3.478505	2200	602.756	17.3925
5.000000	11.000000	2200	602.756	55.0000

80.4280976 mL of primary solution n°1 are needed (16.0856195 mg of IMD).

2.2 D-1 - Preparation of chemical, soils and horse dung

Preparation of the primary solution :

- 20 mg of IMD is weighted in a small cup,
- It is introduced in a 100 mL volumetric flask with small volumes of milliQ water,
- The volumetric flask is filled,
- An ultrasound bath is used to completely dissolve IMD (around 20 min).

Preparation of soils : (water used = Volvic)

- For the controls :
 - 2200 g of soil is weighted in an inox recipient,
 - 602.8 mL of water are added (do not mix),
 - The recipient is covered with aluminum.

- For each concentration
 - 1100 g of dry soil is weighted in an inox recipient twice,
 - Recipient 1 is humidified with 301.4 mL of water,
 - For recipient 2 :
 - * The corresponding V_{prim} is used to create a final solution of 100 mL of water,
 - * This solution is added drop by drop with a glass pipette on the surface of the soil,
 - * The rest of the water (total water added = 301.4 mL) is added on the surface of the soil.
 - Recipients are covered with aluminum.

Preparation of horse dung :

- 450 g of dry horse dung are prepared with 1590 mL of water,
- 45×13.6 g of this preparation is weighted in small boxes for the growth cosms,
- 45×27.2 g of this preparation is weighted in small boxes for the reproduction cosms.

2.3 D0 - Preparation of cosms and earthworms introduction

For each concentration :

- Recipient 1 and recipient 2 are mixed individually and then together for at least 10 min,
- This preparation is used to prepare 10 cosms (254.8 g),
- The rest of this preparation is put into sample pots that are kept in a thermoregulated chamber at -20°C ,
- In 5 cosm (growth), 13.6 g of horse dung is added and mixed,
- In 5 cosm (reproduction), 27.2 g of horse dung is added and mixed.

Earthworms introduction :

- For growth, earthworms are weighted individually and put in their respective cosms,
- For reproduction, earthworms are weighted in groups of 2 and put in their respective cosms,
- Cosms are weighted and their weight is written on their labels to be able to follow the soil humidity,
- Cosms are kept in a thermoregulated enclosure at 18°C .

2.4 Maintenance

On D7, D14 and D21, soil humidity is checked.

On D14, earthworms are fed and weighted the same way as D0. Cosms are weighted again.

2.5 D28 - End of exposition

On D28, earthworms are weighted and put on wet filter paper in Petri dishes.

2.6 D31 - Freeze of individuals and cocoons recovery

Earthworms are weighted and put in test tubes in a -80°C refrigerator.

2.7 D31-56 - Cocoons

Each day of the working week, cocoons eclosions are checked.